

Day 2: Python Comments and Variables

Python Comments

What are comments?

Comments are notes written inside code to explain what the program does. Python ignores comments while executing the program.

Types of Comments

- Single-line comment: Starts with #

Example:

```
# This is a comment  
print("Hello")
```

- Multi-line comments:

Python does not have a dedicated multi-line comment syntax. Developers commonly use multiple # lines or triple quotes (''' ''' or """ """) as documentation strings (docstrings).

Variables

What is a variable?

A variable is a named container used to store data.

Creating Variables

```
x = 10  
name = "Sneha"  
pi = 3.14  
is_student = True
```

Python automatically determines the data type, so no type declaration is needed.

Variable Naming Rules

- Must start with a letter or underscore (_).
- Cannot start with a number.
- Can contain letters, numbers, and underscores.
- Case-sensitive (age and Age are different).
- Cannot use Python keywords (if, for, class, etc.).

Data Types

int – Integer (10)
float – Decimal (3.14)
str – Text ("Hello")
bool – True or False

Examples

```
age = 20  
name = "Alice"  
height = 5.6  
is_coder = True
```

```
print(age)  
print(name)
```

Key Points

- Variables store data.
- Comments improve readability.
- Python is dynamically typed.
- Use meaningful variable names.